

Name: Mahmoud Adel Khalil

ID: 202210183

=====

TASK 1 – DATASET IDENTIFICATION

Dataset 1: NYC Yellow Taxi Trip Data (2023)

- Dataset Name: 2023 Yellow Taxi Trip Data
- Data Source: data.gov (NYC Open Data)

LINK: <https://catalog.data.gov/dataset/2023-yellow-taxi-trip-data>

- Number of Records: ~38 million rows (year 2023 alone)
- Data Format: CSV

Dataset 2: Sentiment140 – Twitter Sentiment Dataset

- Dataset Name: Sentiment140 Dataset with 1.6 Million Tweets
- Data Source: Kaggle

LINK: <https://www.kaggle.com/datasets/kazanova/sentiment140>

- Number of Records: 1,600,000 tweets
- Data Format: CSV

Dataset 3: Global Weather Repository

- Dataset Name: Global Weather Repository (Daily Updating)
- Data Source: Kaggle

LINK: <https://www.kaggle.com/datasets/nelgiriyeWithana/global-weather-repository>

- Number of Records: Millions of daily weather observations across 200+ countries
- Data Format: CSV

TASK 2 – ANALYZE THE FIVE Vs

Dataset	Volume	Velocity	Variety	Veracity	Value
NYC Taxi Trips (2023)	Very High – ~38 million rows for 2023 alone; billions across all years; data in terabytes	High – New trip records logged in real-time as each ride ends. Monthly batch releases also provided	Moderate – Structured CSV with numeric, datetime, and geographic (lat/lon) fields	High – Government-collected, mostly clean and validated; some GPS outliers	High – Enables city planning, traffic analysis, route optimization, and fare pricing models
Sentiment140 (Twitter)	High – 1.6 million tweets; full platform reaches petabytes daily globally	Very High – ~500 million tweets per day globally (~6,000 per second)	High – Includes unstructured text, emoticons, hashtags, URLs, mentions, and metadata fields	Low-to-Moderate Contains spam, sarcasm, slang, bots, and mislabeled sentiments	High – Useful for brand monitoring, public opinion mining, and NLP model training
Global Weather Repository	High – Millions of daily readings across 200+ countries and thousands of stations	Very High – Updated daily from weather stations and satellites worldwide	High – Includes temp, humidity, wind speed, pressure, precipitation, UV & air quality	High – Sourced from official weather stations and satellites; minimal data gaps	Very High – Climate research, disaster fore-casting, agriculture, and aviation safety planning

Detailed Explanation of Each V:

VOLUME - Size of the Dataset:

Volume refers to the sheer amount of data generated and stored.

- NYC Taxi dataset contains ~38 million records for 2023 alone. Across all years since 2009, it amounts to billions of rows, requiring distributed storage systems such as Hadoop or cloud-based data lakes to manage efficiently.
- Sentiment140 holds 1.6 million tweets, representing a sample of a platform that produces hundreds of terabytes of data globally every single day.
- Global Weather Repository aggregates millions of daily meteorological readings across more than 200 countries and thousands of weather stations, making it extremely large in scale when viewed historically.

VELOCITY - Speed at Which Data is Generated:

Velocity describes how quickly new data arrives and must be processed.

- Twitter (Sentiment140) leads all three in velocity. Tweets are generated at roughly 6,000 per second globally, demanding real-time streaming pipelines such as Apache Kafka or Apache Spark.
- Global Weather data is updated daily from thousands of satellite and ground stations simultaneously, requiring continuous ingestion pipelines.
- NYC Taxi data is logged in near real-time per trip but officially released in monthly batches, representing a high-velocity source managed in batch form.

VARIETY - Number of Different Data Types:

Variety refers to the diversity of data formats and structures.

- Global Weather Repository exhibits strong variety: numerical readings (temp, pressure, wind), categorical fields (weather condition descriptions), timestamps, and geographic coordinates all appear together in one dataset.
- Twitter Sentiment140 contains a rich mix: unstructured text, emojis, hashtags, @mentions, URLs, polarity labels, and user metadata.
- NYC Taxi data is primarily structured (numeric fares, datetime pickups/dropoffs, zone IDs), making it the least varied of the three, though still multi-field.

VERACITY - Data Quality and Reliability:

Veracity measures the trustworthiness and accuracy of the data.

- Global Weather Repository has the highest veracity as readings come from calibrated official weather instruments and satellite systems, though occasional sensor failures or missing data gaps exist.

- NYC Taxi data has high veracity as a government-operated collection system, though some GPS errors and fare anomalies are present.
- Sentiment140 has the lowest veracity due to the nature of social media: sarcasm, irony, slang, bots, and mislabeled sentiments all reduce accuracy significantly.

VALUE - Usefulness of the Data:

Value reflects the actionable insights that can be extracted from the data.

- All three datasets have high practical value:
 - * NYC Taxi data informs urban mobility policies, surge pricing algorithms, and city infrastructure investment decisions.
 - * Sentiment140 is used by companies and researchers for brand reputation tracking, election forecasting, and real-time crisis monitoring.
 - * Global Weather data drives life-critical applications including natural disaster forecasting, agricultural planning, aviation safety, and climate change research.

TASK 3 - DISCUSSION QUESTIONS

Q1. Which dataset best represents Big Data?

The NYC Yellow Taxi Trip Dataset best represents Big Data as a whole.

It excels across nearly all Five Vs simultaneously:

- Volume: ~38 million records in 2023 alone; billions across all years.
- Velocity: Continuous, real-time trip logging throughout the day and night.
- Variety: Combines GPS coordinates, timestamps, numeric fare values, passenger counts, and categorical fields (payment type, zone ID).
- Veracity: Government-sourced data with robust automated collection systems.
- Value: Directly applicable to traffic planning, route optimization pricing strategies, and urban policy decisions.

While the Global Weather Repository competes closely in terms of volume and veracity, the NYC Taxi dataset stands out because it originates from a fully automated, always-on data pipeline covering an entire megacity's mobility patterns. Processing this dataset at scale is impossible without distributed computing infrastructure.

Q2. Which of the Five Vs was most evident in your datasets?

VELOCITY was the most dramatically evident characteristic across all three datasets.

- Twitter generates approximately 500 million tweets per day globally - roughly 6,000 per second - requiring real-time streaming pipelines.
- Global Weather systems update readings from thousands of stations every hour, 24 hours a day, across the entire planet simultaneously.
- NYC Taxi trips are logged automatically the moment each ride concludes,

generating a continuous, uninterrupted stream of transactional records.

VOLUME was a close second, as all three datasets contain millions to billions of records - far exceeding what traditional relational databases can handle without specialized infrastructure.

VARIETY also stood out, particularly in the Twitter and Weather datasets, where structured, unstructured, and semi-structured data coexist.

Q3. Why might traditional databases struggle with these datasets?

Traditional relational databases (e.g., MySQL, Microsoft SQL Server) were designed for structured, transactional data at a manageable scale. They struggle with Big Data for several key reasons:

1. VOLUME LIMITATIONS:

Traditional databases store data on a single server or small cluster.

Handling tens of millions of rows (like the Taxi dataset) causes severe performance degradation and storage bottlenecks. Distributed systems like Apache Hadoop HDFS or Amazon S3 are required instead.

2. VELOCITY CONSTRAINTS:

RDBMS systems use ACID transactions (Atomicity, Consistency, Isolation, Durability) which ensure data integrity but are too slow for real-time ingestion at the speed of Twitter's or weather networks' data streams. Streaming platforms like Apache Kafka and Apache Spark Streaming are built specifically for this purpose.

3. VARIETY INCOMPATIBILITY:

Traditional databases use rigid schemas - every row must conform to predefined columns and data types. Unstructured data like tweet text and embedded JSON objects cannot be stored natively without heavy preprocessing. NoSQL databases (MongoDB, Cassandra) and data lakes handle schema-flexible data far more effectively.

4. VERACITY HANDLING:

Traditional systems assume clean, pre-validated input. Big Data sources contain noise, duplicates, null values, and inconsistencies that require dedicated data quality pipelines (e.g., Apache Spark, Python Pandas) before any meaningful analysis can take place.

5. VALUE EXTRACTION:

Extracting real value from Big Data requires machine learning models, natural language processing, and large-scale analytics - capabilities that go far beyond the SQL queries that traditional databases are optimized for.